MSAS – Assignment #1: Simulation

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1 Implicit equations

Exercise 1

Consider the physical model of a mechanical system made of four rigid rods $\mathbf{a}_i$ with $i \in [1;4]$ in a kinematic chain, as shown in Fig. 1. The system has only two degrees of freedom. The objective is to determine, fixing the angle β, the corresponding angle of equilibrium α between the rods $\mathbf{a}_1$ and $\mathbf{a}_2$. Writing the kinematic closure, it is possible to obtain the following equation

$$\frac{a_1}{a_2}\cos\beta - \frac{a_1}{a_4}\cos\alpha - \cos(\beta-\alpha) = -\frac{a_1^2 + a_2^2 - a_3^2 + a_4^2}{2a_2a_4} \tag{1}$$

where a_i is the length of the corresponding rod. This equation, called the equation of Freudenstein, can be clearly re-written in the form $f(\alpha) = 0$ and solved, once β is fixed, for a particular value of α. Assume that $a_1 = 10$ cm, $a_2 = 13$ cm, $a_3 = 8$ cm, and $a_4 = 10$ cm and

1) Implement a general-purpose Newton solver (NS).
2) Solve Eq. (1) with NS for $\beta \in [0, \frac{2}{3}\pi]$ with the analytical derivative of $f(\alpha)$ and tolerance set to 10^{-5}. For every β value, use two initial guesses, -0.1 and $\frac{2}{3}\pi$. What do you observe? You can validate your results with the Matlab© built-in function `fzero`.
3) Repeat point 2) using the derivative estimated through finite differences and compare the accuracy of the two methods.
4) Eq. (1) can be solved analytically only for specific values of β. Find at least one and compare the analytical solution with yours.
5) Repeat point 2) extending the β interval up to π. What do you observe?

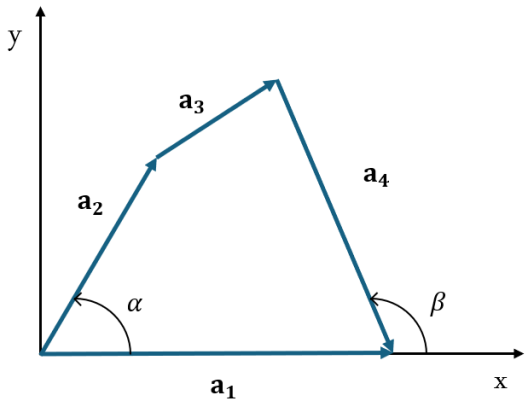


Figure 1: Kinematic chain.

(4 points)

Newton's method is a zero-finding method useful for non-linear equations. It is an iterative method in which the correction at each step is found by using Equation 2, and the solution is found when $|f(x_k)| < tol$.

$$x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)} \tag{2}$$

After the computation of the analytic derivative with respect to α, Equation 3, the Equation 1 has been solved using two different initial guesses. The solutions obtained with NS have been compared against the ones obtained with the Matlab© built-in function `fzero`.

$$\frac{df(\alpha)}{d\alpha} = \frac{a_1}{a_4} sin(\alpha) - sin(\beta - \alpha) \tag{3}$$

Figure 2 shows the solutions for both the initial guesses, it is important to notice how the solution changes depending on the initial guess.

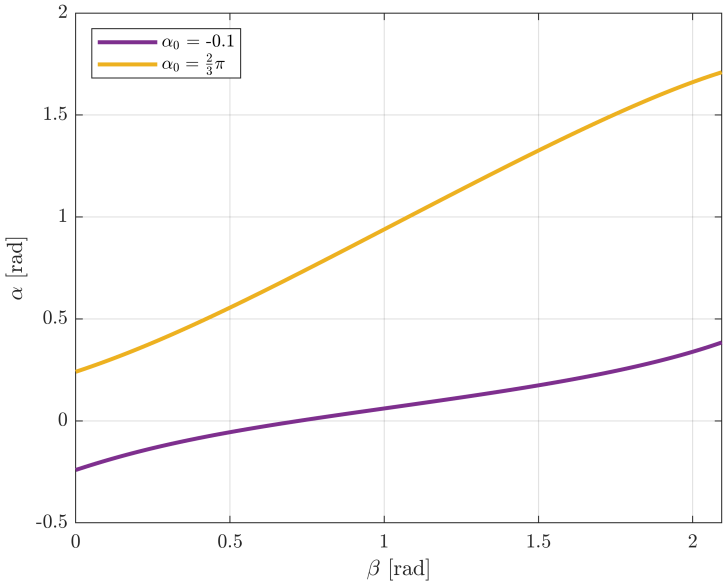


Figure 2: Solution obtained with NS with analytic derivative

Figure 3 shows how the solutions are close to each other, in fact, the error ranges between $[1e-9;\ 1e-5]\,rad$.

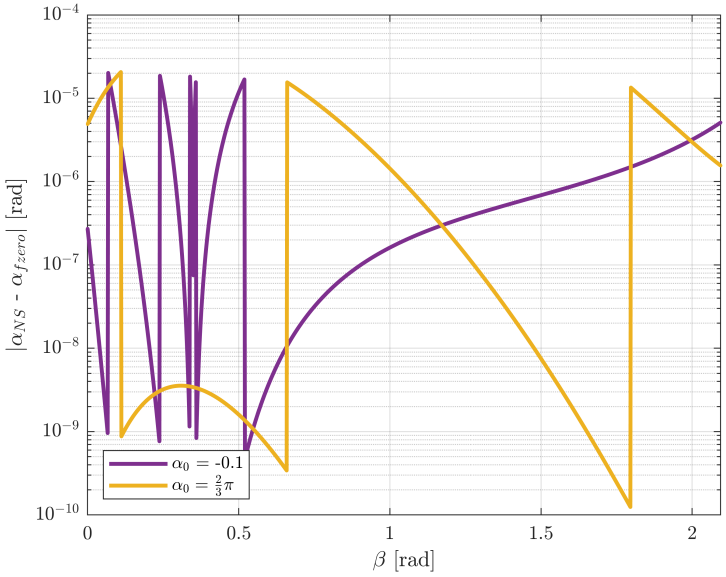


Figure 3: Error between NS with analytical derivative and `fzero`

The same procedure has been executed using the NS with the derivative approximated through central finite differences, Equation 4, it has been preferred to backward and forward finite

differences for its higher accuracy.

$$f'(x) \simeq \frac{f(x+\epsilon) - f(x-\epsilon)}{2\epsilon} \tag{4}$$

The size of the perturbation, ϵ, in Equation 2 has been computed with Equation 5.

$$\epsilon = \begin{cases} |x|\sqrt{\epsilon} & \text{if } |x| \geq 1 \\ \sqrt{\epsilon} & \text{if } |x| < 1 \end{cases} \tag{5}$$

Figure 4 illustrates how the two solutions are similar, in fact, the difference is in the order of $1e-12\,rad$.

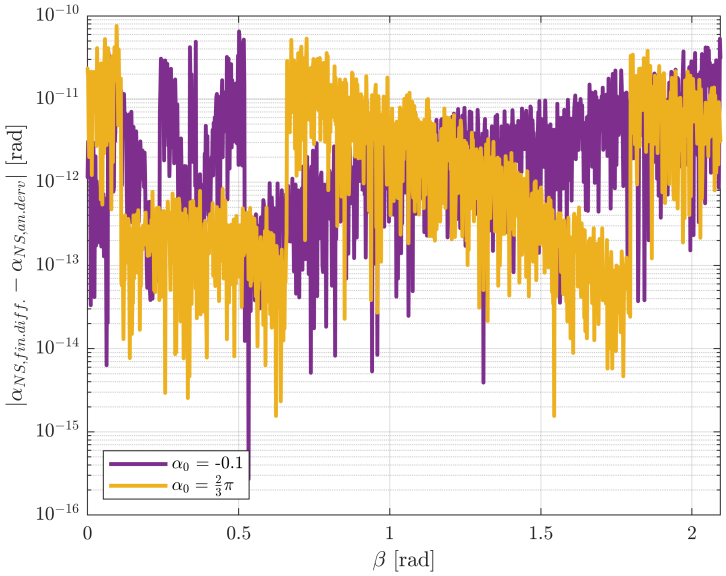


Figure 4: Error between NS with analytical derivative and NS with central finite differences

An analytical solution of Equation 1 can be found for $\beta = 0,\ \pi/2,\ \pi$.

- $\beta = 0$, the analytical solution has been computed as follows:

$$\left(\frac{a_1}{a_2} + \frac{a_1^2+a_2^2-a_3^2+a_4^2}{2a_2a_4}\right) + \left(-\frac{a_1}{a_4} - 1\right) cos(\alpha) = 0 \implies \frac{101}{104} = cos(\alpha)$$
$$\implies \alpha = \begin{cases} -0.2408\,rad \\ 0.2408\,rad \end{cases} \tag{6}$$

- $\beta = \pi/2$, the computing of the analytical solution has been achieved by applying some trigonometric formulas:

$$\left(\frac{a_1^2+a_2^2-a_3^2+a_4^2}{2a_2a_4}\right) - \frac{a_1}{a_4}cos(\alpha) - sin(\alpha) = 0 \implies \frac{61}{52} = cos(\alpha) + sin(\alpha)$$
$$\implies \frac{61}{52\sqrt{2}} = cos(\alpha - \tfrac{\pi}{4}) \implies \alpha = \begin{cases} 0.1928\,rad \\ 1.3780\,rad \end{cases} \tag{7}$$

- $\beta = \pi$, this value of beta leads to the following equation, which is an impossible equality, therefore it does not have any solution. The NS is not able to find any solution, it runs until the maximum number of iterations is reached.

$$\left(\frac{a_1}{a_2} + \frac{a_1^2+a_2^2-a_3^2+a_4^2}{2a_2a_4}\right) + \left(-\frac{a_1}{a_4} + 1\right) cos(\alpha) = 0 \implies \frac{21}{52} = 0 \tag{8}$$

As shown in Equation 8 for $\beta = \pi$ there is no solution, this happens because of the geometrical configuration of the kinematic chain, this leads to a lock-up of the system before β reaches π. In Figure 5 and Figure 6, it is possible to observe how the behavior of the system, for β higher than approximately 2.6 rad completely diverges.

β	$\lvert\alpha_{analytical} - \alpha_{NS}\rvert$ [rad]
0	$2.7324e-07$
$\pi/2$	$8.1302e-07$

(a) $\alpha_0 = -0.1$

β	$\lvert\alpha_{analytical} - \alpha_{NS}\rvert$ [rad]
0	$4.8746e-06$
$\pi/2$	$3.0098e-09$

(b) $\alpha_0 = 2\pi/3$

Table 1: Error between analytical solution and the one computed with NS

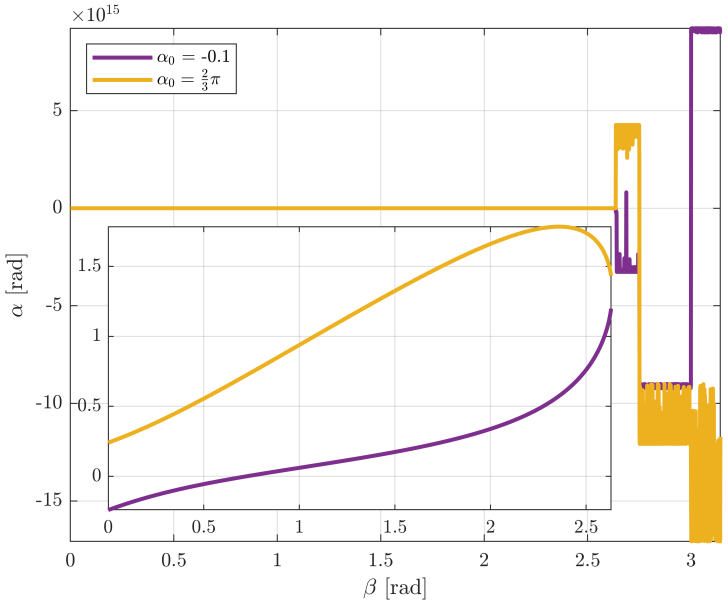


Figure 5: Solution obtained with NS with analytic derivative for $\beta \in [0, \pi]$

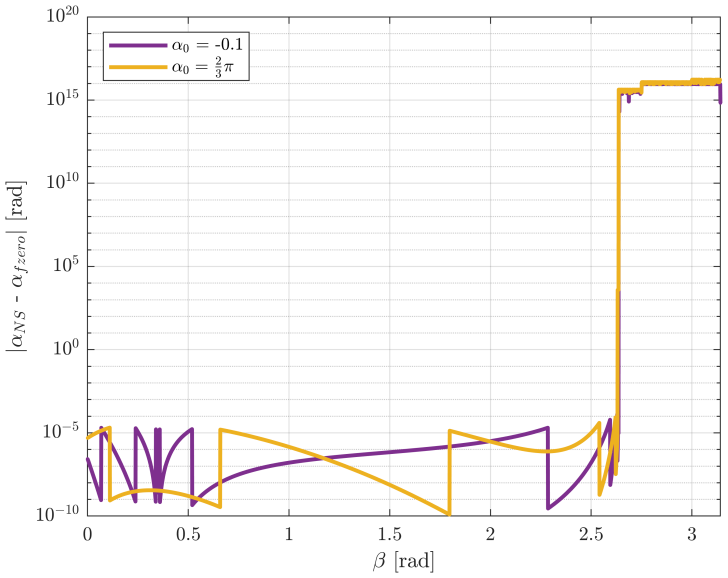


Figure 6: Error between NS with analytical derivative and `fzero` for $\beta \in [0, \pi]$

2 Numerical solution of ODE

Exercise 2

The physical model of a moving object with decreasing mass is shown in Fig. 7. The mathematical model of the system is expressed by the equations:

$$m(t)\frac{dv}{dt} = \overline{F} + f(t) - \alpha \cdot v \tag{9}$$

$$m(t) = m_0 - c_m \cdot t \tag{10}$$

$$\overline{F} = -\,0.5 \cdot \rho \cdot C_d \cdot A_m \cdot v^2 \tag{11}$$

Assuming that: $v(0) = 0$ [m/s], $m_0 = 20$ [kg], $c_m = 0.1$ [kg/s], $f(t) = 1$ [N], $\alpha = 0.01$ [Ns/m], $\rho = 0$ [kg/m^3], and Eq. (9) has the exact solution:

$$v(t) = \frac{f(t)}{\alpha} - \left[\frac{f(t)}{\alpha} - v(0)\right]\left[1 - \frac{c_m \cdot t}{m_0}\right]^{\frac{\alpha}{c_m}} \tag{12}$$

Answer to the following tasks:

1) Implement a general-purpose, fixed-step Heun's method (RK2);
2) Solve Eq. (9) using RK2 in $t \in [0, 160]$s for $h_1 = 50$, $h_2 = 20$, $h_3 = 10$, $h_4 = 1$ and compare the numerical vs the analytical solution;
3) Repeat points 1)–2) with RK4;
4) Trade off between CPU time & integration error.

Now, assuming that: the fluid density is $\rho = 900$ [kg/m^3], $C_d = 2.05$, and $A_m = 1$ [m^2], answer to the following tasks:

1) Solve Eq. (9) using RK2 in $t \in [0, 160]$s for $h_1 = 1$;
2) From the results of the previous point, use a proper ode of Matlab to solve Eq. (9) ;
3) Discuss the results.

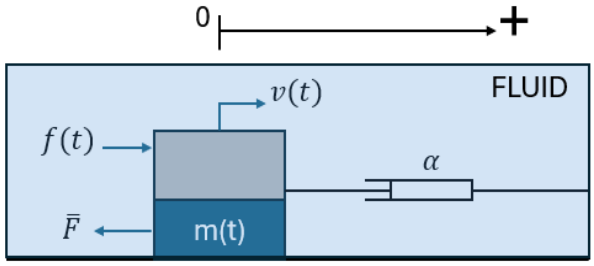


Figure 7: Moving object with decreasing mass.

(5 points)

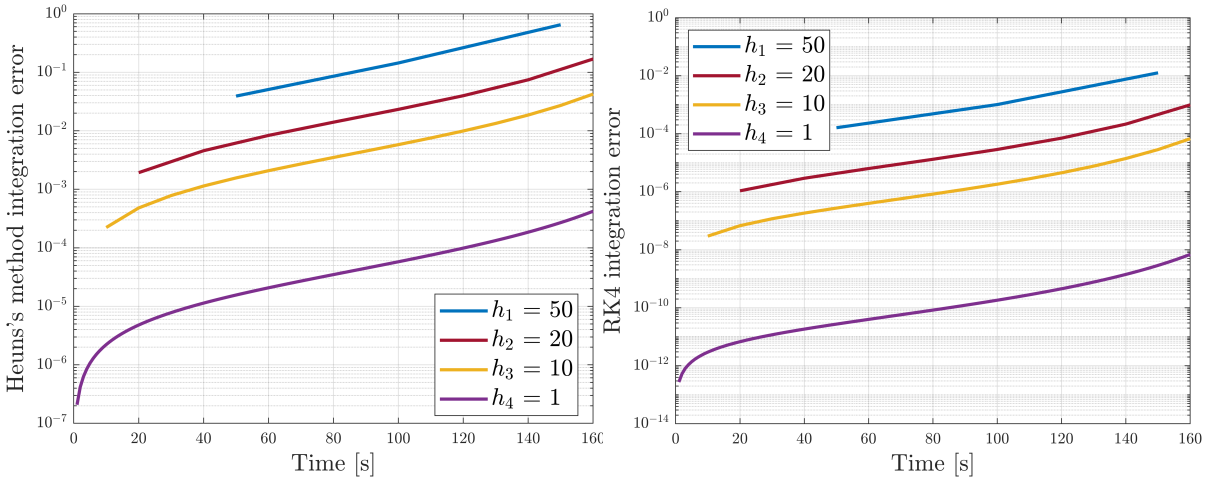
Heun's method is a second-order method of the Runge-Kutta family, its generic formulation is presented in Equation 13.

$$\begin{aligned} \mathbf{x}_{k+1}^P &= \mathbf{x}_k + h\mathbf{f}(\mathbf{x}_k,\, t_k) \\ \mathbf{x}_{k+1} &= \mathbf{x}_k + \frac{h}{2}\mathbf{f}(\mathbf{x}_k,\, t_k) + \frac{h}{2}\mathbf{f}(\mathbf{x}_{k,+1}^P\, t_{k+1}) \end{aligned} \tag{13}$$

Runge-Kutta 4 is a fourth-order method, whose generic formulation is reported in Equation 14

$$\begin{aligned}
\mathbf{x}^{P1} &= \mathbf{x}_k + \frac{h}{2}\mathbf{f}(\mathbf{x}_k,\, t_k) \\
\mathbf{x}^{P2} &= \mathbf{x}_k + \frac{h}{2}\mathbf{f}(\mathbf{x}^{P1},\, t_k + \frac{h}{2}) \\
\mathbf{x}^{P3} &= \mathbf{x}_k + h\mathbf{f}(\mathbf{x}^{P2},\, t_k + \frac{h}{2}) \\
\mathbf{x}_{k+1} &= \mathbf{x}_k + \frac{h}{6}\mathbf{f}(\mathbf{x}_k,\, t_k) + \frac{h}{3}\mathbf{f}(\mathbf{x}^{P1},\, t_k + \frac{h}{2}) + \frac{h}{3}\mathbf{f}(\mathbf{x}^{P2},\, t_k + \frac{h}{2}) + \frac{h}{6}\mathbf{f}(\mathbf{x}^{P3},\, t_k + h)
\end{aligned} \tag{14}$$

Figure 8 shows the difference between the analytical solution and one obtained with the two methods for different step size values h.



(a) Heuns's method **(b)** RK4

Figure 8: Integration error

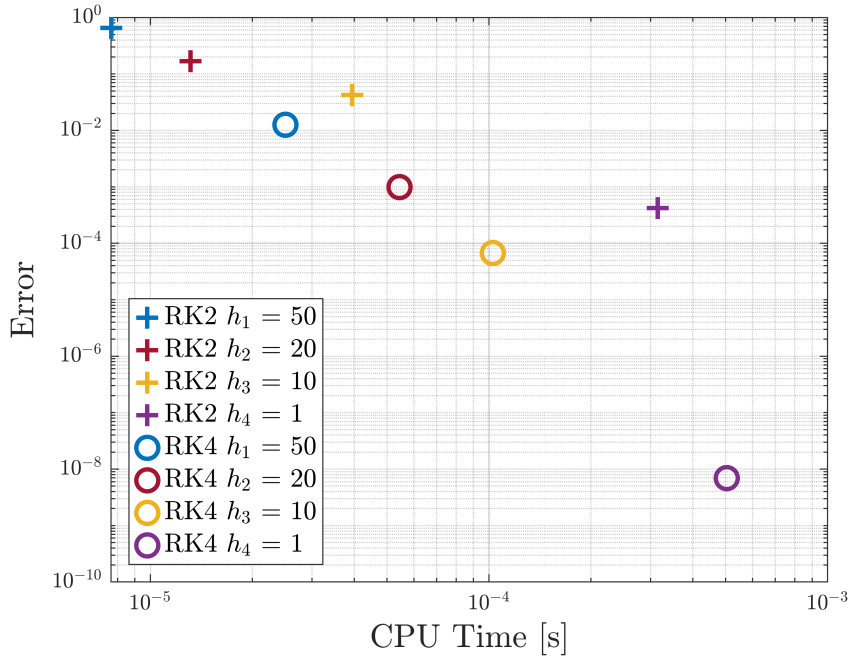


Figure 9: CPU time against error for RK2 and RK4, for the different values of h

The error increases for larger values of h, this is because the approximation of the solution is

rougher. By analyzing Figure 9, it is possible to observe how for the same value of h, RK4 is more accurate than Heun's, due to its higher-order. Another comparison can be made on the time required to run the algorithm, it has been computed using the Matlab© built-in function `timeit`. The results in Figure 9 show as one might expect, that an accurate solution requires more computational time. The selection of the method is a trade-off between accuracy and computational time, for those problems in which high accuracy is not mandatory, RK2 represents a good choice, RK4 with $h = 10$ is a good compromise between error and CPU time.

After the change in the value of the fluid density, it can be observed from Figure 10 that the solution of RK2 completely diverges from the exact solution, computed using `ode23s`. This happened because the step size used was higher than the maximum one allowed. The selection of the ode has been made by considering that the system has no high tolerance, so a simple `ode23` can be used, but it provides a solution with 397 steps, so an ode solver for stiff problems has been analyzed, in particularly `ode15s` and `ode23s`, which provide a solution with 42 and 29 steps respectively, considering that `ode23s` has been selected.

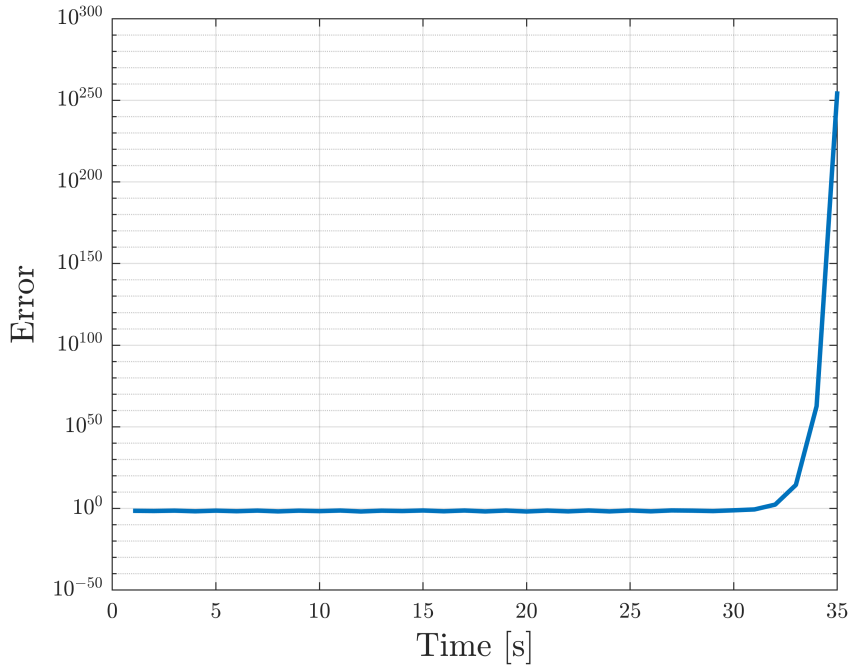


Figure 10: Error between RK2 with $h = 1$ and the exact solution computed with *ode23s*

The maximum step size can be calculated by computing the eigenvalue, λ, and then substituting it into Equation 15.

$$h_{max}\lambda = -2 \tag{15}$$

To compute λ first is necessary to linearize Equation 11 around the equilibrium velocity, found by solving Equation 9 equal to zero, the equilibrium velocity is $v_{eq} = 0.0329\, m/s$. After F has been linearized using the Taylor series, then it has been replaced in Equation 9.
By computing λ and substituting it into Equation 15, $h_{max} \approx 0.13$ is obtained.

Exercise 3

Let $\dot{\mathbf{x}} = A(\alpha)\mathbf{x}$ be a two-dimensional system with $A(\alpha) = [0, 1; -1, 2\cos\alpha]$. Notice that $A(\alpha)$ has a pair of complex conjugate eigenvalues on the unit circle; α denotes the angle from the Re$\{\lambda\}$-axis.

a) Runge-Kutta methods

1) Write the operator $F_{\text{RK2}}(h, \alpha)$ that maps $\mathbf{x}_k$ into $\mathbf{x}_{k+1}$, namely $\mathbf{x}_{k+1} = F_{\text{RK2}}(h, \alpha)\,\mathbf{x}_k$. 2) With $\alpha = \pi$, solve the problem "Find $h \geq 0$ s.t.max $(|\text{eig}(F(h, \alpha))|) = 1$". 3) Repeat point 2) for $\alpha \in [0, \pi]$ and draw the solutions in the $(h\lambda)$-plane. 4) Repeat points 1)–3) with RK4.

b) Backinterpolation methods

Consider the backinterpolation method $\text{BI2}_{0.3}$. 1) Derive the expression of the linear operator $B_{\text{BI2}_{0.3}}(h, \alpha)$ such that $\mathbf{x}_{k+1} = B_{\text{BI2}_{0.3}}(h, \alpha)\mathbf{x}_k$. 2) Using the same approach of a), draw the stability domain of $\text{BI2}_{0.3}$ in the $(h\lambda)$-plane. 3) Derive the domain of numerical stability of BI2_θ for the values of $\theta = [0.2,\ 0.4,\ 0.6,\ 0.8]$.

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a) Runge-Kutta methods

The operator $F_{RK2}(h,\ \alpha)$ has been obtained by substituting f with $A(\alpha)$ in Equation 13, after all the simplification the expression of the operator is represented in Equation 16.

$$\mathbf{F}_{RK2}(h,\ \alpha) = \mathbf{I} + h\mathbf{A}(\alpha) + \frac{1}{2}h^2\mathbf{A}^2(\alpha) \tag{16}$$

The same procedure has been used to compute $F_{RK4}(h,\ \alpha)$ operator shown in Equation 17.

$$\mathbf{F}_{RK4}(h,\ \alpha) = \mathbf{I} + h\mathbf{A}(\alpha) + \frac{1}{2}h^2\mathbf{A}^2(\alpha) + \frac{1}{6}h^3\mathbf{A}^3(\alpha) + \frac{1}{24}h^4\mathbf{A}^4(\alpha) \tag{17}$$

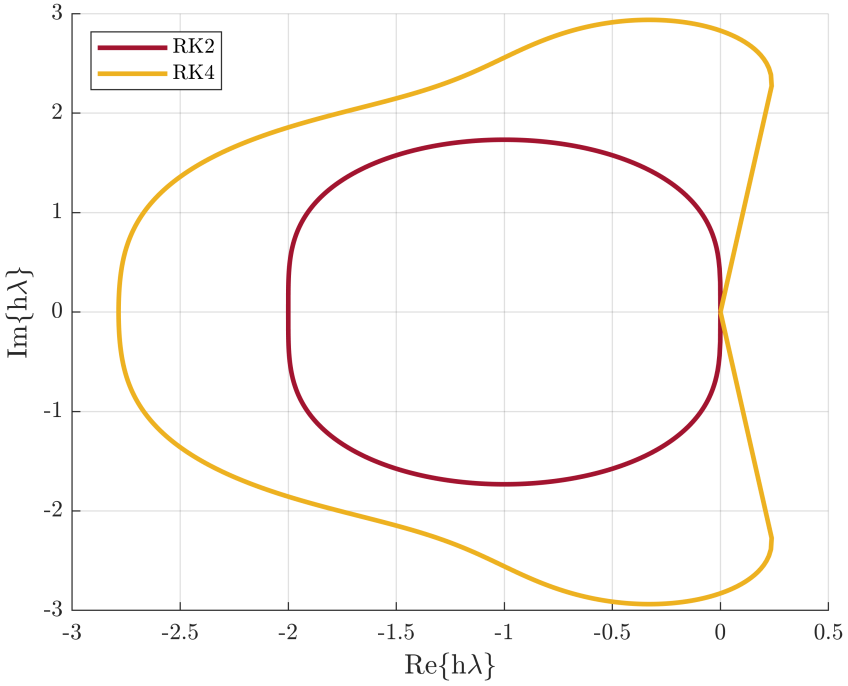


Figure 11: Stability region for Runge-Kutta 2 and Runge-Kutta 4

The value of the step sizes, h for $\alpha = \pi$ are reported in Table 2, they have been computed by solving "Find $h \geq 0$ s.t.max $(|\text{eig}(F(h, \alpha))|) = 1$" by using `fzero` with an initial guess close to the solution.

Operator	h
$\mathbf{F}_{RK2}$	2.0000
$\mathbf{F}_{RK4}$	2.7853

Table 2: Step size for $\alpha = \pi$

The stability domains have been obtained by using the same procedure explained above by exploiting numerical continuation: the previous value of h has been used as the initial guess for the next value of alpha, and this procedure has been executed iteratively, for α going from π to 0.

b) Backinterpolation methods

Backinterpolation methods are methods composed of two steps:

- an implicit method with a step size equal to θh:

$$\mathbf{x}_{k+\theta h} = \left[\mathbf{I} + \theta h \mathbf{A}(\alpha) + \frac{1}{2}(\theta h)^2 \mathbf{A}^2(\alpha)\right] \mathbf{x}_k \tag{18}$$

- the remaining fraction $(1 - h)\theta$ with implicit method:

$$\mathbf{x}_{k+\theta h} = \left[\mathbf{I} - (1-\theta) h \mathbf{A}(\alpha) + \frac{1}{2}[(1-\theta)h]^2 \mathbf{A}^2(\alpha)\right]^{-1} \mathbf{x}_{k+\theta h} \tag{19}$$

For BI2 both the methods explicit and implicit are second-order. The operator $\mathbf{B}_{BI2_\theta}$ is reported in Equation 20

$$\mathbf{B}_{BI2_\theta}(h, \alpha) = \left[\mathbf{I} - (1-\theta) h \mathbf{A}(\alpha) + \frac{1}{2}[(1-\theta)h]^2 \mathbf{A}^2(\alpha)\right]^{-1} \left[\mathbf{I} + \theta h \mathbf{A}(\alpha) + \frac{1}{2}(\theta h)^2 \mathbf{A}^2(\alpha)\right] \tag{20}$$

As for the RK2 and RK4, the stability of the BI2 region has been found by solving the same problem.

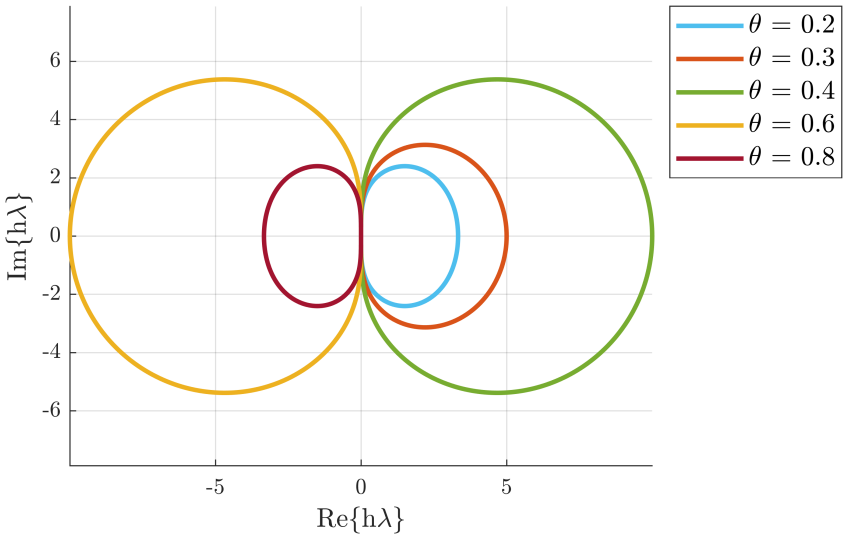


Figure 12: Stability region for $BI2_\theta$ methods

In Figure 12 are represented the stability region for the different values of theta, in the left half plane there are the methods with more explicit behavior, in fact, the stability region is inside the lines, while in the right half plane the motion with more implicit behavior, the region is outside the lines.
It is important to point out that in order to have a correct representation of the stability region, the computation has been done from $\alpha = pi$ to 0 for $\theta = [0.6, 0.8]$, which are the region in the left half plane, while from 0 to π for the other values of θ located in the right half plane. The initial values used are [5, 3, 10, 10, 3] respectively for $\theta \in [0.3, 0.2, 0.4, 0.6, 0.8]$.
The last important consideration that can be made concerns the lines: $\theta = 0.2$ and $\theta = 0.9$ have symmetric curves, it happens because the first one is 20% explicit and 80% implicit, the opposite for the second one. The same situation is verified in $\theta = 0.4$ and $\theta = 0.6$.

Exercise 4

The cooling problem of a high-temperature mass is shown in Fig. 13. Assuming that: K_c= 0.0042 [J/(s K)] and K_r = 6.15 x 10^{-11} [J/(s K^4)] are the convective and radiation heat loss coefficients respectively, $\widetilde{T}$ is the mass temperature in time, T_a = 277 [K] is the surrounding air temperature, C= 45 [J/K] is the mass thermal capacity, and $\widetilde{T}(0)$ = 555 [K] is the initial mass temperature.

1) Write the mathematical model of the system.
2) Solve the mathematical model using: RK2 with h= 720, and RK4 with h= 1440.
3) Compare the solution of point 2) with the "exact" one. Discuss the results. (Hint: use an ODE solver of Matlab).

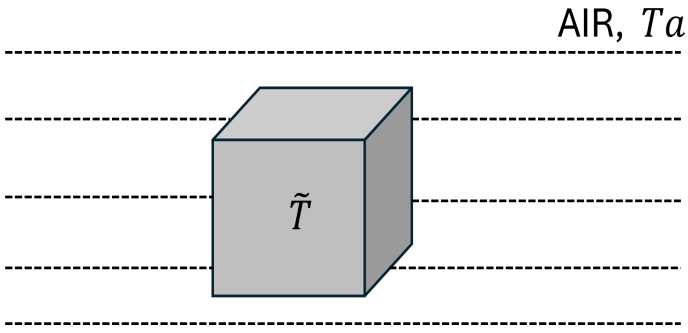


Figure 13: The cooling of high-temperature mass.

(4 points)

Considering the system, the mathematical model is obtained by applying the conservation of energy at the mass.

$$C\frac{d\tilde{T}}{dt} = -Q_{conv} - Q_{rad} \tag{21}$$

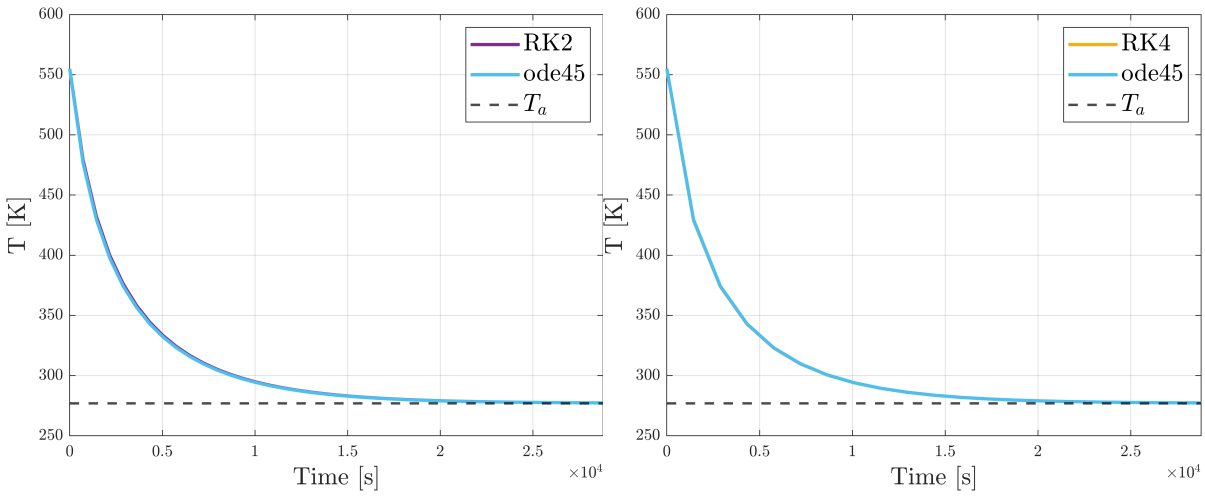
Where Q_{conv} and Q_{rad} are the convective and radiative thermal contributes.

$$Q_{conv} = K_c(\tilde{T} - T_a) \tag{22}$$

$$Q_{rad} = K_r(\tilde{T}^4 - T_a^4) \tag{23}$$

$$\frac{d\tilde{T}}{dt} = -\frac{K_c}{C}(\tilde{T} - T_a) - \frac{K_r}{C}(\tilde{T}^4 - T_a^4) \qquad \tilde{T}(0) = 555\,K \tag{24}$$

Equation 24 has been solved using RK2 and RK4 with two different time steps. For both methods, it has been integrated from $t(0) = 0s$ until the system reaches the asymptotically equilibrium temperature T_a, a $t_f = 20h_{RK4}s$ has been chosen.
Both the solutions have been compared against the exact one obtained using `ode45`. It has been chosen because it has a medium accuracy and the problem is non-stiff.
Figure 14 shows how both the solutions are really close to the exact one, a small difference is noticeable for RK2 solution, while the RK4 one is almost perfectly overlapped with the exact.



(a) Runge-Kutta 2 **(b)** Runge-Kutta 4

Figure 14: Exact solution against the ones computed with Runge-Kutta

By analyzing Figure 15, it is possible to observe how the error for both methods, starts with a maximum and then decreases. As expected the RK4 is more accurate than RK2 despite its double time step.

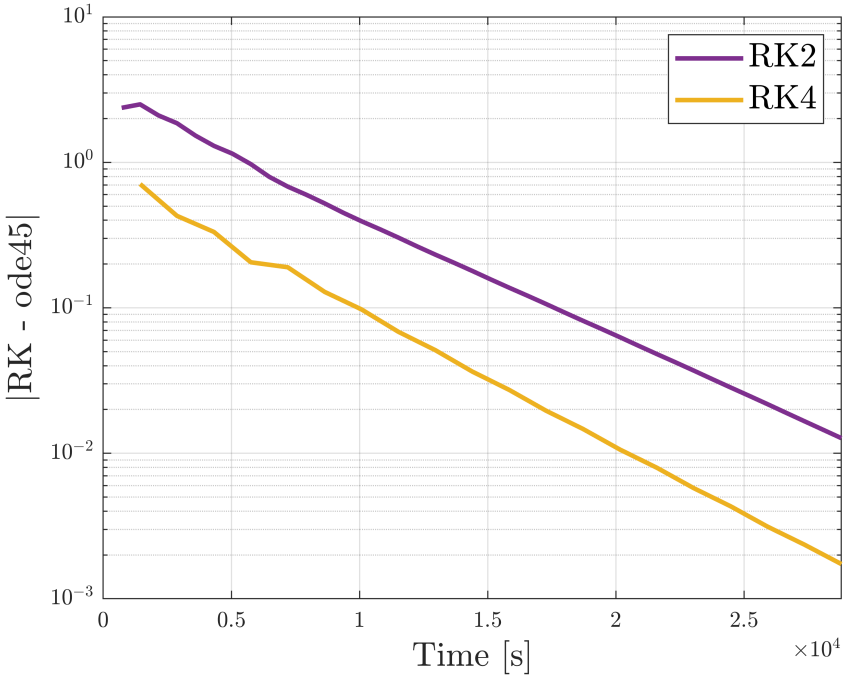


Figure 15: Error between the exact solution and the ones computed with Runge-Kutta

Exercise 5

Consider the electrical circuit in Fig. 16. At time t>=0 the switch disconnects the battery and the capacitor discharges its stored energy to the circuit. Answer to the following tasks:

1) Find the state variable form of the circuit with x_1=q and x_2=$\frac{dq}{dt}$, where q is the charge on capacitor.
2) Show that the system is stiff when the circuit parameters are R=25 [Ω], L = 20 [mH], C = 200 [mF], and v_0 = 12 [V]. Represent the eigenvalues on the ($h\lambda$)-plane both for RK2 and IEX4 stability domain.
3) Use IEX4 to simulate the transient response of the system with the parameters of point 2), and determine the largest step size such that RK2 yields a stable and accurate solution.
4) Discuss the results.

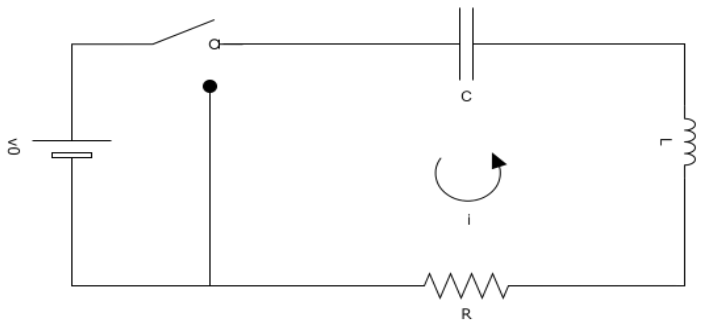


Figure 16: Electric circuit model.

(4 points)

The system the in Figure 16 can be described by the Kirchoff' voltage law, Equation 25.

$$v_0 = v_R + v_L + v_C \tag{25}$$

where v_R, v_L, v_C are the voltage of the resistor, the inductor and the capacitor, respectively.

$$v_R = Ri \quad v_L = L\frac{di}{dt} \quad v_C = \frac{q}{C} \tag{26}$$

By recalling that $i = \dfrac{dq}{dt}$ and by substituting it and Equation 26 into Equation 25, Equation 27 has been obtained.

$$v_0 = L\frac{d^2q}{dt^2} + R\frac{dq}{dt} + \frac{q}{C} \tag{27}$$

Considering the state variables x_1=q and x_2=$\frac{dq}{dt}$, the Homogeneous ODE associated to Equation 27, can be written in state space form.

$$\begin{cases} \dot{x_1} = x_2 & x_1(0) = Cv_0 \\ \dot{x_2} = -\dfrac{1}{LC}x_1 - \dfrac{R}{L}x_2 & x_2(0) = 0 \end{cases} \implies \begin{bmatrix} \dot{x_1} \\ \dot{x_2} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -\dfrac{1}{LC} & -\dfrac{R}{L} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \implies \dot{\mathbf{x}} = \mathbf{A}\mathbf{x} \tag{28}$$

The eigenvalues of **A** show that the system is stiff, due to the fact that differs in 3 orders of magnitude.

λ_1	λ_2
-0.2000	-1249.8

Table 3: Eigenvalues of $\mathbf{A}$

To represent the eigenvalues with the stability domain for RK2 and IEX4 the same procedure explained in Equation 2 has been used. The operator $\mathbf{F}_{IEX4}$ is reported in Equation 29.

$$\mathbf{F}_{IEX4}(h,\,\alpha) = -\frac{1}{6}\left[\mathbf{I} - hA(\alpha)\right]^{-1} + 4\left[\mathbf{I} - \frac{h}{2}A(\alpha)\right]^{-2} - \frac{27}{2}\left[\mathbf{I} - \frac{h}{3}A(\alpha)\right]^{-3} + \frac{32}{3}\left[\mathbf{I} - \frac{h}{4}A(\alpha)\right]^{-4} \tag{29}$$

To represent the eigenvalues in the $h\lambda$ plane is necessary to multiply both for a value of h, for representation purposes only, it has been set equal to 1.
In Figure 17 are represented both the stability region with the eigenvalues, is possible to notice that for IEX4 the stability region is outside the lines, so for every value of h the eigenvalues are in the stability region, while for RK2 the stability region is inside the lines, so λ_1 is inside this region, while λ_2 is outside. Considering that there is a h_{max} which is the maximum step size allowed for RK2 in order to have a stable and accurate solution. The value of h_{max} has been computed such that $h_{max}\lambda_2 = -2$, for this system $h_{max} = 0.0016003$.

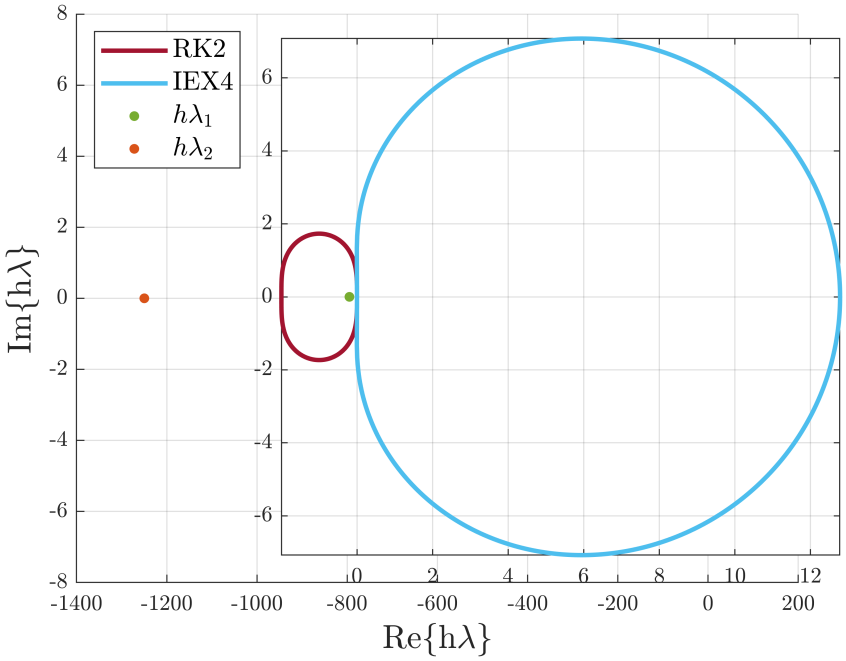
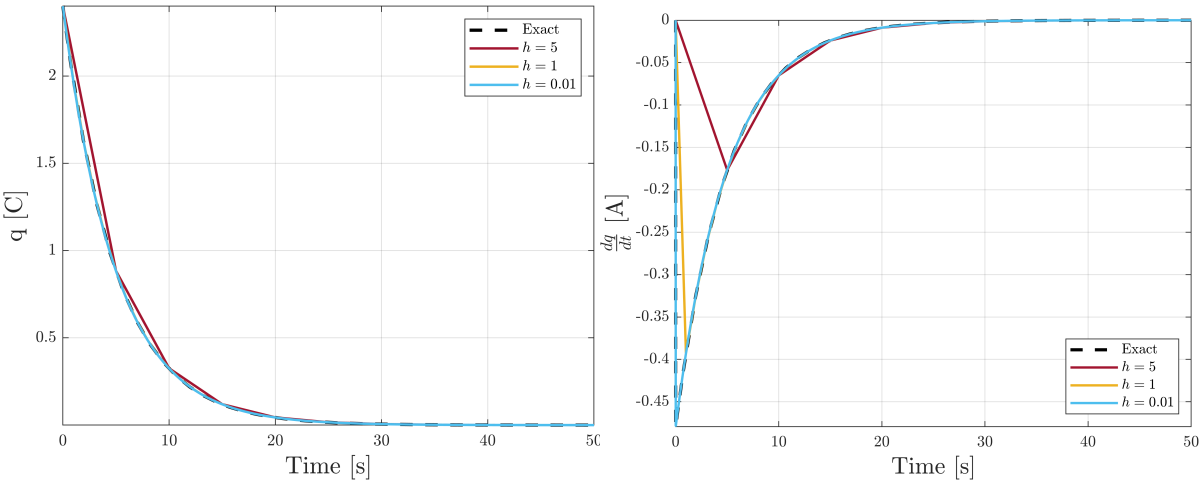


Figure 17: Satbility regions of Runge-Kutta 2 and Implic Extrapolation Technique od Order 4, with the two eigenvalues of the system multiplied by $h = 1$

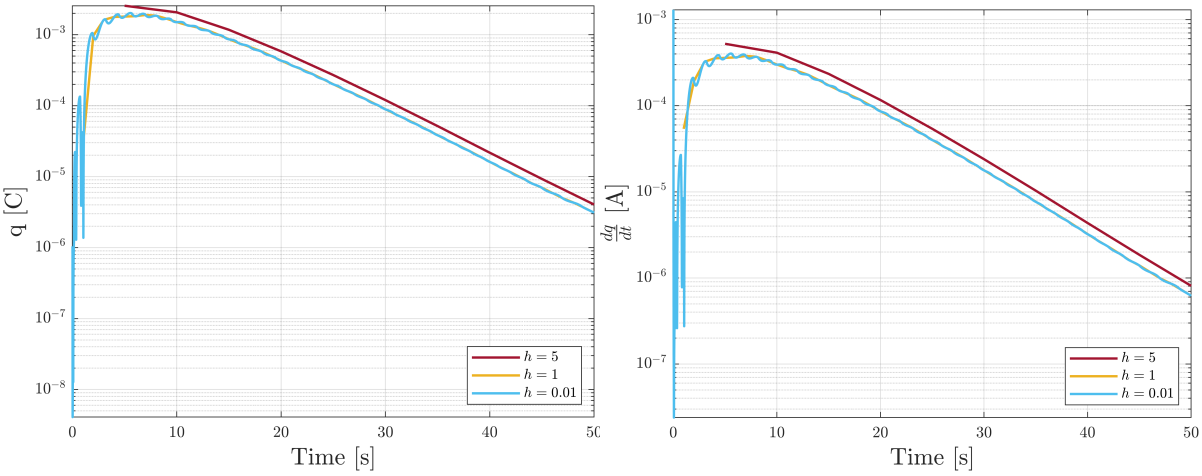
The system has two different time constants $\tau_= - \frac{1}{\lambda_1} = 4.9992s$ and $\tau_= - \frac{1}{\lambda_2} = 0.0008s$, τ_2 is very fast while τ_1 is slower.
The transient response of the system has been simulated from $t(0) = 0s$ to $t_f = 50s$ to have that the transient response is finished, so considering that this happens around $5\tau_2 \approx 25s$, the $50s$ chosen are sufficient.
The solution has been computed using three different step sizes for both methods. For RK2 one is slightly higher than h_{max}, one is equal to h_{max} and the last is smaller than the maximum one.
Figure 18 shows the solutions computed with IEX4 for three different values of $h \in [5,\, 1,\, 0.01]$. The exact solution has been obtained by using the `ode23s`.

By analyzing Figure 18 is possible to observe that the charge q follows the behavior of the ones computed with the ode solver. For the current, the situation is different, in fact, for the higher values of h IEX4 is not capable of representing correctly the fast transient.



(a) Capacitor charge **(b)** Current

Figure 18: Solutions computed with IEX4 with the three values of h against the exact one computed with `ode23s`



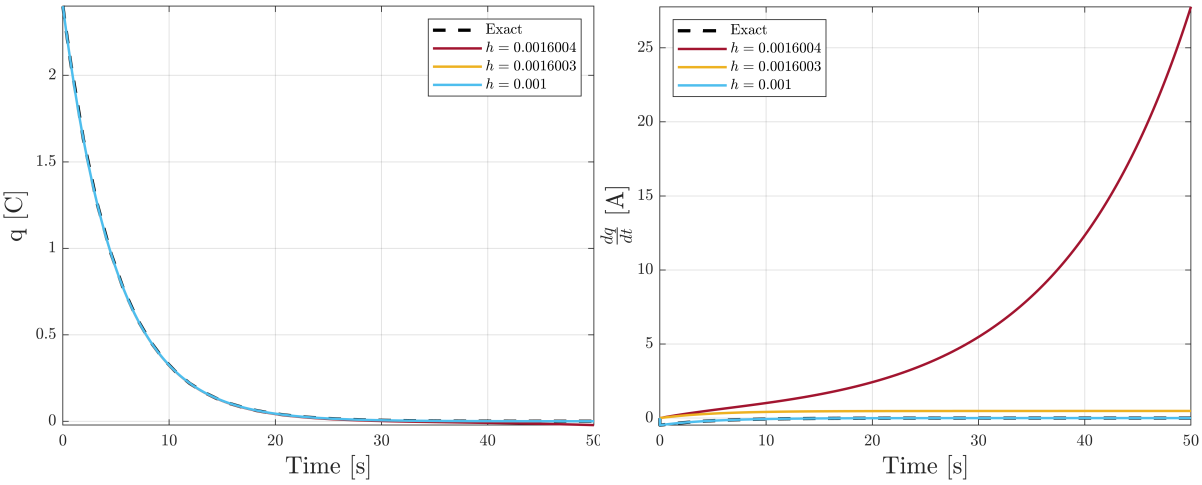
(a) Capacitor charge **(b)** Current

Figure 19: Errors between solutions computed with IEX4 with the three values of h against the exact one computed with `ode23s`

The outcomes in Figure 20 and Figure 21 show a behavior coherent with what has been discussed above, regarding the stability of RK2. For all the three step sizes selected [0.0016004, h_{max}, 0.001], the q is almost correctly represented, only for $h > h_{max}$ it starts to diverge in the last second of the simulation, this can be observed by looking at the error.
For the current the situation is different:

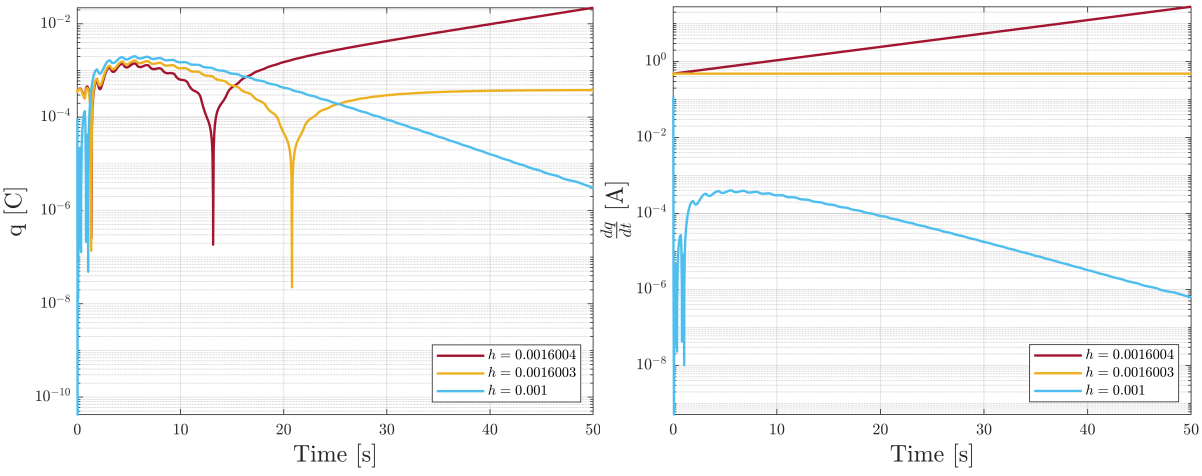
- $h > h_{max}$ it does not follow the exact solution, and the error grows in time;
- $h = h_{max}$ happens that the system is mapped in an asymptotically stable system, so i converges to a value different from zero;
- $h < h_{max}$ it follows correctly the behavior of the exact solution.

In conclusion, for RK2 the step size needs to be selected accurately to have a correct representation of the evolution of the system's state, in fact, also a slight shift from h_{max} can cause a drift from the correct solution.



(a) Capacitor charge **(b)** Current

Figure 20: Solutions computed with RK2 with the three values of h against the exact one computed with `ode23s`



(a) Capacitor charge **(b)** Current

Figure 21: Errors between solutions computed with RK2 with the three values of h against the exact one computed with `ode23s`

Exercise 6

Consider the bouncing ball physical model in Fig. 22. The mathematical model of the system is represented by the following equations:

$$\frac{dv}{dt} = -g + \frac{F_D}{m_b} + \frac{F_A}{m_b} \tag{30}$$

$$\frac{dx}{dt} = v \tag{31}$$

$$v^+ = -k \cdot v^- \tag{32}$$

$$F_D = -\ 0.5 \cdot \rho \cdot C_d \cdot A_b \cdot v \cdot |v| \tag{33}$$

$$F_A = \rho \cdot V_b \cdot g \tag{34}$$

where the initial conditions are: $v(0) = 0$ [m/s], and $x(0) = 10$ [m].

Assuming that: the ball velocity before (v^-) and after (v^+) the ground impact is governed by the attenuation factor $k = 0.9$, the air density is $\rho = 1.225$ [kg/m^3], the ball mass is $m_b = 1$ [kg], the ball area is $A_b = 0.07$ [m^2], the ball volume is $V_b = 0.014$ [m^3], and the drag coefficient is C_d=1.17, answer to the following tasks:

1) Solve the mathematical model in the time interval $t \in [0, 10)$s using an ode integrator of Matlab.
2) Repeat point 1) for ρ=15 [kg/m^3]. Which is the difference with respect to point 1)?
3) Now solve the problem when the fluid density is ρ=60 [kg/m^3] in $t \in [0, 10)$s. Discuss the result.
4) Repeat point 3) using RK4 for $h = 3$. Discuss the result.

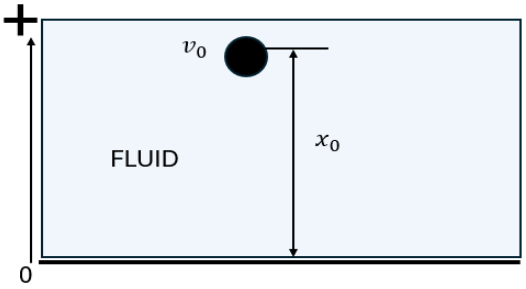
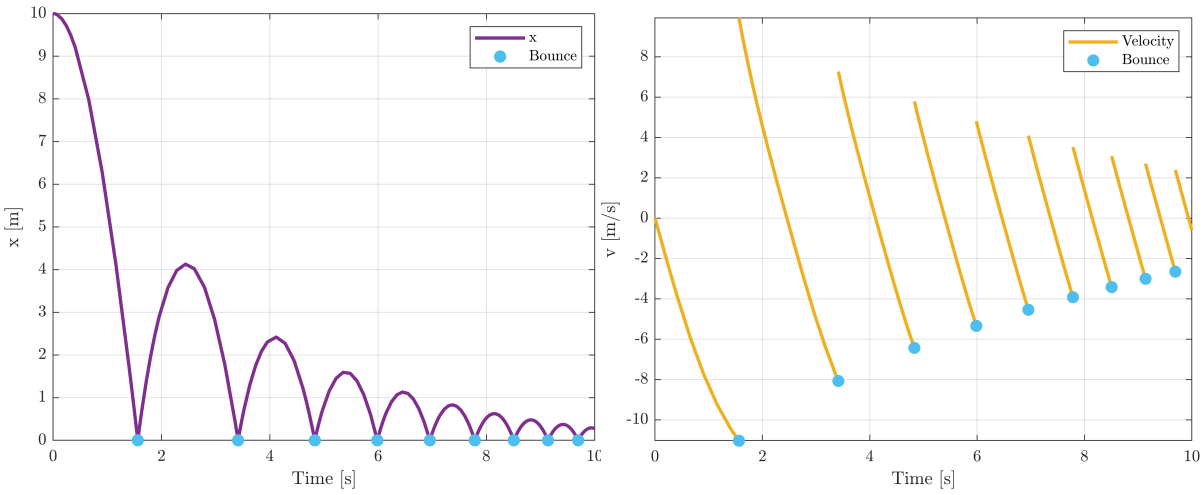


Figure 22: Bouncing ball system.

(5 points)

In order to integrate correctly the Equation 30 and Equation 31 some considerations need to be made. The integrator selected is `ode45`, this integrator is used for non-stiff problems and it has a medium accuracy, which is suitable for this problem.
Then in order to represent the ground impact it is necessary to use the option `'Event'` in the ode setting; this option takes as input an event function which at each step checks if the condition specified inside of it is equal to zero, in this case the condition is the height of the ball, when this condition is triggered the integration stops. After that the new initial condition is composed by the last value of height in the previous integration and the velocity has been set using Equation 32, this procedure is repeated iteratively until t_f is lower than $10s$.
The last consideration regards the height of the bounce, in fact, especially for $\rho = 15\,kg/m^3$ happens that after a series of bounces the ball starts to go below the ground, this is not physically

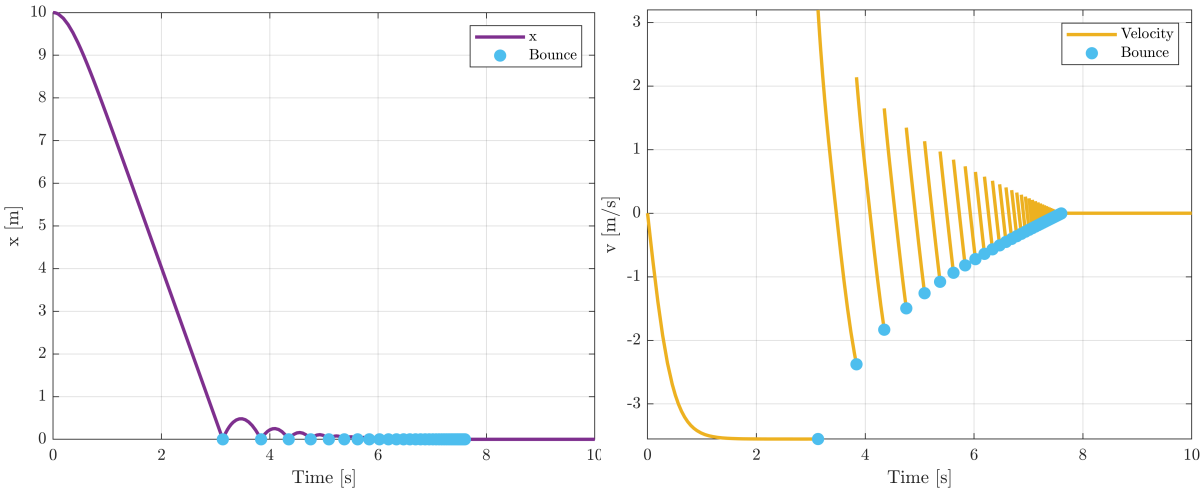
possible, to prevent this type of problems a check on the value of bounce has been made, when the height of this goes below $1e-6\,m$ the velocity and acceleration have been set equal to zero. In Figure 23 is reported the solution with $\rho = 1.225\,kg/m^3$. It is possible to observe that in this case the height of the bounces is above the limit defined above. The discontinuity in the velocity is due to the fact that as defined above the new initial condition for the velocity has been set with Equation 32.



(a) Position **(b)** Velocity

Figure 23: Position and velocity with $\rho = 1.225\,kg/m^3$

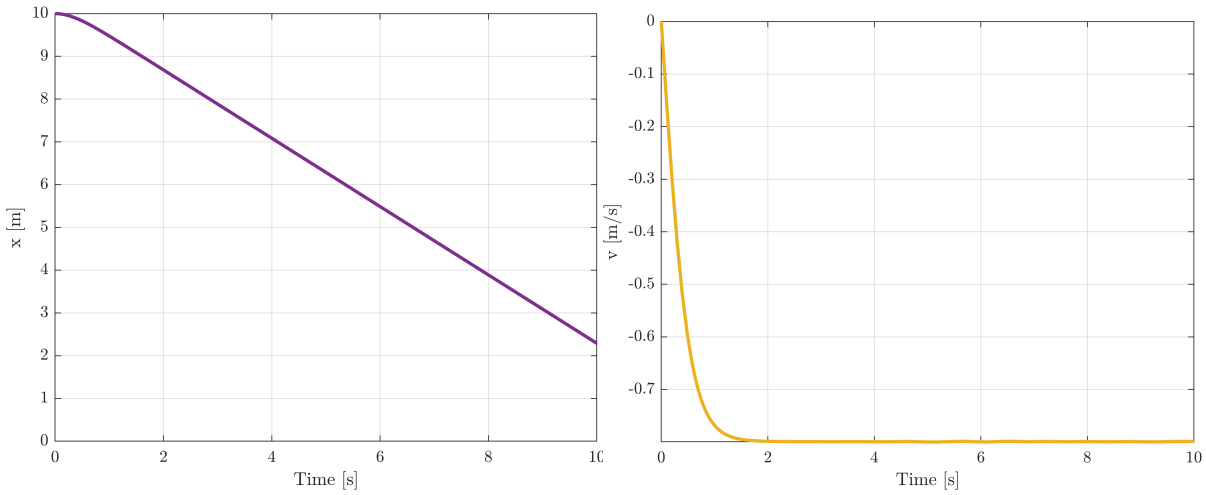
With the increment of the density up to $\rho = 15\,kg/m^3$ both the forces give a higher contribution, also the first impact arrives later with respect to the first case before the first bounce is possible to notice that around $2\,s$ the acceleration goes to zero due to fact that the velocity becomes constant. Another peculiarity is that around $8\,s$ the ball stops bouncing in fact, both position and velocity go to zero.



(a) Position **(b)** Velocity

Figure 24: Position and velocity with $\rho = 15\,kg/m^3$

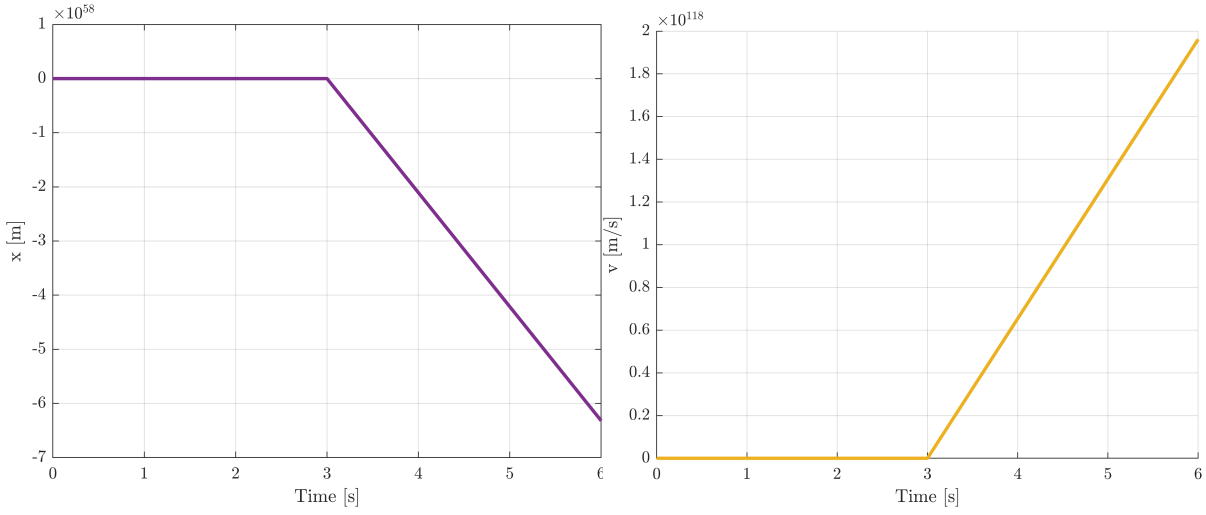
For $\rho = 60\,kg/m^3$, happens what has been discussed above, the acceleration goes to zero around $2\,s$ so the velocity is constant and the position follows a linear evolution, and in $10\,s$ it has not been reached the ground.



(a) Position **(b)** Velocity

Figure 25: Position and velocity with $\rho = 60\,kg/m^3$

Figure 26 shows the solution obtained using RK4 with $h = 3$, as shown with this time step the solution diverges from the exact one because the step selected is too high.



(a) Position **(b)** Velocity

Figure 26: Position and velocity with $\rho = 60\,kg/m^3$, solution obtained with RK4, for $h = 3$

To obtain an estimation of h_{max} the same procedure explained in section 2 has been applied:

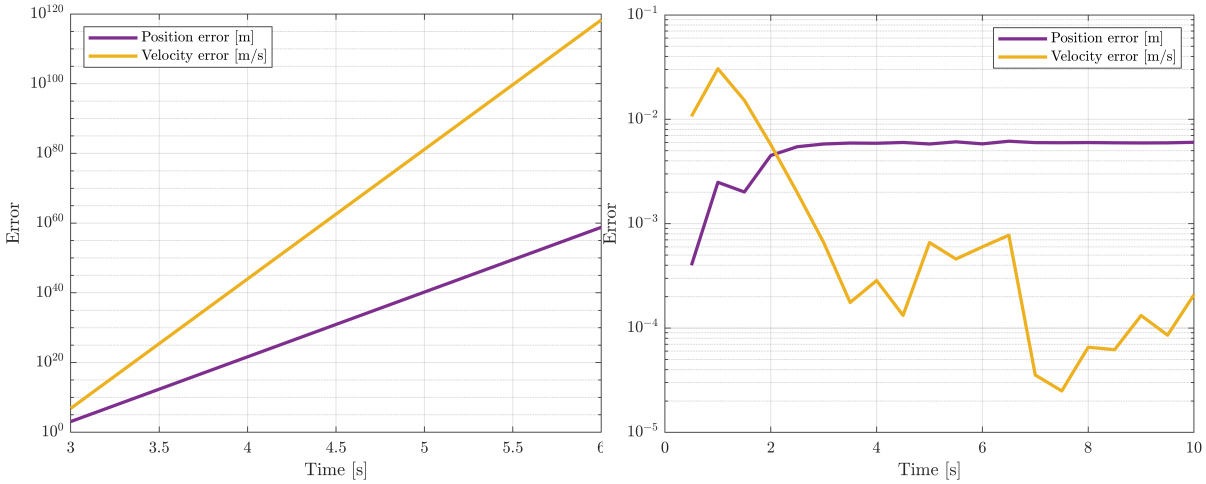
- determination of equilibrium velocity;
- linearization through Taylor expansion of F_D, around the equilibrium velocity;
- write the equation in a state space representation;
- computation of the eigenvalues.

Then the h_{max} has been computed by using Equation 35, the value is $h_{max} = 0.7092$.

$$h_{max}\lambda = -2.7853 \tag{35}$$

After the determination of h_{max} another integration with RK4 has been performed but with $h = 0.5$, as shown in Figure 27 the error with this time step is around $1e-2$ for the position

and $1e-4$ for the velocity while for $h = 3$ they are approximately in the order $1e60$ for the position and $1e120$ for the velocity.



(a) Runge-Kutta 4 with $h = 3$

(b) Runge-Kutta 4 with $h = 0.5$

Figure 27: Errors between solutions computed with RK4 with the two values of h and the exact one computed with `ode45`